

Eugene Henninger-Voss

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EDUCATION

Ph.D in Mathematics - Expected graduation May 2027

2022-Present

The Pennsylvania State University

State College, PA

- PhD in Mathematics under Jack Huizenga studying Algebraic Geometry

B.S. of Mathematics - Minors in German and Linguistics

2017-2021

Tufts University

Medford, MA

- Summa cum laude with honors in thesis (*Mathematical Aspects of Moran's I*)

SELECTED RESEARCH AND TEACHING POSITIONS

Graduate Instructor

2022-Present

Pennsylvania State University

State College, PA

- Primary Instructor (6 semesters): College Algebra II; Calculus I; Math of Music
- Assistant Course Coordinator (2 semesters): College Algebra II
 - Coordinated instructors, lecture content and course materials; oversaw grading; mentored new instructors; wrote exams
- Accessibility Project Team (Spring 2026)
 - Wrote Python scripts and AI agents (Claude Code) to convert files into more accessible formats to comply with updated ADA regulations

Research and Development Engineer Intern

Summer 2026

Applied Research Laboratory

State College, PA

- Used mathematics to analyze and guide development of machine learning
- Developed theory for, implemented in Python, and rigorously assessed improvements
- Doubled validation speed; improved training speed by 30%; decreased validation loss by 2/3

Research Associate

Winter 2020, 2021-2022

Keystone Research Center

Harrisburg, PA

- Built model of state funding of K-12 education and related state taxes and rebates
 - Modeled the impacts of proposed policies on students, schools, and taxpayers
 - Numerically optimized policy proposals with Python
- Created estimates of wealth and the effect of wealth taxes using income tax data
- Presented models and policy recommendations to stakeholders, including the governor of PA

Research Fellow

Summer 2019

Moravian Research Experience for Undergraduates

Bethlehem, PA

- Investigated open problems in graph theory using traditional and experimental mathematics

Research Fellow

Summer 2018

Voting Rights Data Institute

Somerville, MA

- Interdisciplinary research to measure and detect gerrymandering
- Wrote Python programs investigating graph theory, Markov chain Monte-Carlo algorithms, and more

SKILLS

- *Natural Languages*: Portuguese (conversational); German (conversational)
- *Computer Languages & Systems*: Python (incl. NumPy, SciPy, PyTorch); C++; Git
- *Technical*: Machine learning (classical and neural networks); Data analysis; Modeling
- *Non-technical*: Technical communication, Leadership, Organizing